

N7.4

Дано:

$$T_{N_2} = 373 \text{ K}$$

$$\bar{v}_{\text{ср. и в } O_2} = \bar{v}_{\text{ср. и в } N_2}$$

Найти: T_{O_2}

Решение:

$$\sqrt{\frac{3RT_{N_2}}{M_{N_2}}} = \sqrt{\frac{3RT_{O_2}}{M_{O_2}}} \Rightarrow T_{O_2} = M_{O_2} \frac{T_{N_2}}{M_{N_2}}$$

$$= \frac{0,032 \cdot 373}{0,028} = 426 \text{ K.}$$

N7.16

Решение:

$$1) dP = \underbrace{\left(\frac{m}{2\pi kT} \right) \exp\left(-\frac{m\sigma^2}{2kT}\right) 2\pi\sigma d\sigma}_{q(\sigma)}$$

$$2) \frac{dP}{d\sigma} = \frac{m}{2\pi kT} \left(\cancel{2\pi \exp(\quad)} - \cancel{2\pi\sigma \cdot \frac{m \cdot 2\sigma}{2kT} \exp(\quad)} \right) = 0$$

$$1 - \frac{m\sigma^2}{kT} = 0 \Rightarrow \sigma_{\text{н.б.}} = \sqrt{\frac{kT}{m}}$$

$$3) \bar{v}_{\text{ср. и в.}} = \sqrt{\int_0^{+\infty} \sigma^2 dP} = \sqrt{\int_0^{+\infty} \sigma^2 q(\sigma) d\sigma}$$

$$\int_0^{+\infty} \sigma^2 \cdot \left(\quad \right) \exp\left(-\frac{m\sigma^2}{2kT}\right) \cdot 2\pi\sigma d\sigma =$$

$$= \frac{m}{2\pi kT} \cdot 2\pi \int_0^{+\infty} \sigma^3 \exp\left(-\frac{m\sigma^2}{2kT}\right) d\sigma =$$

$$= \frac{m}{kT} \cdot \frac{1}{2 \cdot \left(\frac{m}{2kT} \right)^2} = \frac{m}{kT} \cdot \frac{4k^2 T^2}{2m^2} = \frac{2kT}{m}$$

$$v_{\text{ср. кв}} = \sqrt{\frac{2kT}{m}}$$

$$4) \frac{v_{\text{н.б}}}{v_{\text{ср. кв}}} = \frac{1}{\sqrt{2}}$$

№ 7.18

Дано: Решение:

R, N, T

Найти:

1) $j(t)$ - ?

j, t_0, v_0

$$j(t) = \frac{dN}{S dt} = \frac{1}{4\pi r^2} \cdot \frac{dN}{dt}$$

$$dP = \left(\frac{m}{2\pi kT} \right)^{\frac{3}{2}} \exp\left(-\frac{mv^2}{2kT}\right) \cdot 4\pi v^2 dv = \frac{dn}{n_0} = \frac{dN}{N_0}$$

$$j(t) = \frac{1}{4\pi r^2} \left(\frac{m}{2\pi kT} \right)^{\frac{3}{2}} \exp\left(-\frac{mv^2}{2kT}\right) \cdot 4\pi v^2 N_0 \cdot \frac{dv}{dt}$$

$$= \frac{N_0 v^2}{r^2} \left(\right)^{\frac{3}{2}} \exp(\) \frac{dv}{dt}$$

$$2) v = \frac{R}{t}; \quad v = \frac{r}{t}; \quad \frac{dv}{dt} = -\frac{r}{t^2}$$

$$\therefore j(t) = \frac{N_0}{r^2} \cdot \cancel{v^2} \left(\right)^{\frac{3}{2}} \exp(\) \cdot -\frac{r}{t^2}$$

$$j(t) = \frac{N_0 r}{t^4} \left(\frac{m}{2\pi kT} \right)^{\frac{3}{2}} \exp\left(-\frac{m r^2}{2kT t^2} \right)$$

$$3) j'(t) = \left(\frac{m}{2\pi kT} \right)^{\frac{3}{2}} \left(-\frac{4 N_0 r}{t^5} \exp\left(-\frac{m r^2}{2kT t^2} \right) + \frac{N_0 r}{t^4} \cdot \frac{2 m r^2}{2kT t^3} \right) = 0$$

$$\frac{4}{t} = \frac{m r^2}{kT t^3} \Rightarrow t = \frac{r}{2} \sqrt{\frac{m}{kT}} = t_{\max}$$

$$4) v_{\max} = \frac{r}{t_{\max}} = 2 \sqrt{\frac{kT}{m}}$$